A

R Arginine
V Valine
F Phenylalanine
I Isoleucine
G Glycine
S Serine
M Methionine
E Glutamate
W Tryptophan
K Lysine
A Alanine
Y Tyrosine
P Proline
L Leucine
H Histidine
D Aspartate
C Cysteine
T Threonine
Q Glutamine
N Asparagine

p=0.0002 *** (BH=0.0029)
p=0.016 * (BH=0.053)
p=0.012 * (BH=0.053)
p=0.081 ns (BH=0.15)
p=0.18 ns (BH=0.27)
p=0.39 ns (BH=0.52)
p=0.028 * (BH=0.07)
p=0.7 ns (BH=0.74)
p=0.41 ns (BH=0.52)
p=0.86 ns (BH=0.86)
p=0.49 ns (BH=0.57)
p=0.16 ns (BH=0.26)
p=0.61 ns (BH=0.68)
p=0.41 ns (BH=0.52)
p=0.014 * (BH=0.053)
p=0.061 ns (BH=0.14)
p=0.069 ns (BH=0.14)
p=0.019 * (BH=0.053)
p=0.01 * (BH=0.053)
p=0.00029 *** (BH=0.0029)

Property
Aromatic
Negatively charged
Nonpolar aliphatic
Polar uncharged
Positively charged
Special cases

−0.004 −0.003 −0.002 −0.001 0.000 0.001 0.002 0.003 0.004
Mixed-model β (P/LP − gnomAD) for ΔFull amino-acid fraction

B

Special cases
Positively charged
Polar uncharged
Nonpolar aliphatic
Negatively charged
Aromatic

Σβ=-0.0004 (1 aa)
Σβ=+0.0030 (3 aa)
Σβ=-0.0044 (5 aa)
Σβ=+0.0020 (6 aa)
Σβ=-0.0004 (2 aa)
Σβ=+0.0004 (3 aa)

−0.0100 −0.0075 −0.0050 −0.0025 0.0000 0.0025 0.0050 0.0075 0.0100
Σ β over amino acids in class

C

pLDDT (metapredict)
Frac. R+K
Frac. positive
Frac. basic
Frac. charged (FCR)
Mean net charge
Kappa (charge patterning)
Frac. aromatic
Frac. disorder-promoting
Helix propensity
Frac. acidic
Frac. negative
Disorder (metapredict)

▲ q=0.000 ***
▲ q=0.000 ***
▲ q=0.000 ***
▲ q=0.004 **
▲ q=0.004 **
▲ q=0.051 trend
▲ q=0.038 *
▲ q=0.427 n.s.
▲ q=0.438 n.s.
▼ q=0.850 n.s.
▼ q=0.741 n.s.
▼ q=0.741 n.s.
▼ q=0.000 ***

−0.4 −0.2 0.0 0.2 0.4 0.6
Mixed-model β (P/LP − gnomAD), SD units